

Jack Bosh Part 5

Q1) LLaMA-2 13B param calculation

We have $(3DF + 2D(N+k)H + O(n)) \cdot 2 + DV$

$$\text{So, param} = ((3 \cdot 5,120 \cdot 13,824) + (2 \cdot 5120 \cdot (80) \cdot 128) + O(5120))L + (5120)(32,000)$$

$$= 12,852,019,200 \approx 13e9 \text{ ie 13 billion}$$

$$\begin{array}{r} 1.3 \\ \times 40 \\ \hline 0.00 \\ 42.00 \\ \hline 42.00 \end{array}$$

Q2) B5 = 16M tokens w/ADAM

Per token, we have [5120] ie 10240 bytes. We checkpoint after each PFW matmul. So per layer, we store $L(2(2(B \times F)) + (B \times D))$ bytes = $4(16M \cdot 13824) + 2(16M \cdot 5120)$
 $\approx 1.0512 \text{ bytes or } 1.05 \text{ Terabytes for activations}$
 $= 42 \text{ TB}$

13 billion params at 64b = 26 billion bytes = $26e9 \text{ bytes} = 26 \text{ Gigabytes}$

Per parameter we store 24 bytes of optima state, so we have $8 \cdot 13 \text{ billion} = 104 \text{ GB}$.

So we have $42.13 \text{ TB} + 104 \text{ GB} = 42.13 \text{ TB}$

Q3) context = 32k tokens w/ total batch size of 3M on TPUv5p 16x16x16 slice, bfloat16 weights + float32 optimizer

1) We cannot fit parameters into HBM, NO (TPU v5p HBM, 96GB, but params + optimizer state are 130GB).

2) ~~We should be~~ Since we efficiently split optim state + weights across data parallel shards, and ~~only~~ gather + reduce as needed, we ~~have~~ can definitely fit params + optim state on chip (130/4096) $\rightarrow \approx 10.25GB$

We store $2 \cdot L \cdot (2BF + 130) = 42TB$ but we can shard 4096 ways, so we have $42/4096$ ie 10.25GB. So if we do pure FSP, we can comfortably ~~store~~ do it!

However, we won't be compute bound since the per device batch size is $732 < 840$

$$3) \text{ ~~We should~~ } X_{opt} = \sqrt{\frac{B}{F} \frac{M_x}{M_y}} N = \sqrt{\frac{36 \cdot 40}{13824 \cdot 4096}} \sqrt{\frac{M_x}{M_y}}$$

$$X \approx 512$$

$$\rightarrow X_{opt} = 666 \quad (\text{for } M_x=1, M_y=2) = 942 \sqrt{\frac{M_x}{M_y}}$$

$$\text{or } X_{opt} = 1332 \quad (\text{for } M_x=2, M_y=1)$$

$$\downarrow X \approx 1024$$

If $X=512$ (in the case that $M_x=1, M_y=2$),
we have that to ensure we are compute bound, we want

$$\frac{B}{N} > \frac{\alpha^2}{M_x M_y F} \therefore \text{we will be compute bound for the same batch size regardless of whether we give } M_x=1 \text{ or } M_y=1 \text{ (the other being set to 2 as is required for a 3D mesh)}$$

This because if we dedicate an additional mesh axis to FSP or P, we scale the ~~number of chips we want~~ must scale the corresponding X/Y proportionally to balance the eq.

$$\max \left(\frac{F}{Y \cdot M_x}, \frac{B}{X \cdot M_y} \right)$$

Imagine we have X_{opt}, Y_{opt} so that the two are equal. let $M_x=1, M_y=2$

~~if $M_x=2, M_y=1$~~

~~$\frac{F}{2Y^*} = \frac{B}{X^*}$~~ This holds now let say we want to solve for the following

~~$$\frac{F}{Y} = \frac{B}{2X} \iff \frac{2F}{2Y} = \frac{B}{X} \iff \frac{F}{Y} = \frac{B}{X}$$~~

~~$$\frac{F}{Y} = \frac{B}{2X}$$~~

We want to find X, Y so that the eq holds.

we know $\frac{F}{2Y^*} = \frac{B}{X^*}$ X^* Y^* $M_x=2$ $M_y=1$

let $Y = 2Y^*$ or $X = \frac{X^*}{2}$

then plugging in, we see $\frac{F}{2Y^*} = \frac{B}{X^*}$, which we know is true. So, it turns out

Is this should mean t_{comm} is independent of whether $M_x=1$ or $M_y=1$.

- ▷ (since for their respective optima differ by factors of M_x, M_y , so the net work out. The t_{comm} for their respective optima are the same.

so, lets take $X \approx 512$.

so we first note t_{comm} We are compute bound when

$$\frac{B}{N} \rightarrow \frac{2550}{13284 \cdot 2} \quad \text{or when } \frac{B}{N} \approx 0.235$$

Take $X_{opt} \approx 512$ then we do 512 way FSDP and 8 way TP

Mem per chip = ~~130GB~~ 130GB ~~active~~ activ, sharded 512.8, so

$$\frac{130GB}{4096}$$

* - This will be the same as for the pure FSDP case since $X \cdot Y = 4096$, but they will be sharded over X and Y .

$$FLOPs = \frac{(6 \cdot 3M \cdot 13B \cdot 13B)}{4.59e4 \cdot 4096} = \frac{5880.12 \cdot 10}{4} = 0.31 \text{ seconds}$$

↓ 40% utilization

~~over~~
FLOPs/s

~~over~~ parallelisation